Michael Lulic

Results

Q1)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    int numOfItems;
    double priceItem, taxRate;
    double subtotal, taxAmount, total;

    cout << "Enter the number of items: ";
    cin >> numOfItems;

    cout << "Enter the price per item: ";
    cin >> priceItem;

    cout << "Enter the tax rate (%): ";
    cin >> taxRate;

    subtotal = numOfItems * priceItem;
    taxAmount = subtotal * (taxRate/100);
    total = subtotal + taxAmount;

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    cout << "Subtotal: $" << subtotal << endl;
    cout << "Tax: $" << taxAmount << endl;
    cout << "Total: $" << total << endl;

    return 0;
}
```

```
Enter the number of items: 3
Enter the price per item: 20
Enter the tax rate (%): 15
Subtotal: $60.00
Tax: $9.00
Total: $69.00
```

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Q2)

```cpp
#include <iostream>

using namespace std;

int main(){
    int secondsEntered, sec, min, hour;
    double minAsDouble;

    cout << "Enter the duration in seconds: ";
    cin >> secondsEntered;

    sec = secondsEntered;

    hour = sec / (60*60);
    sec = sec % (60*60);

    min = sec / 60;
    sec = sec % 60;

    minAsDouble = (double) secondsEntered / 60;

    cout << "Hours: " << hour << endl;
    cout << "Minutes: " << min << endl;
    cout << "Seconds: " << sec << endl;
    cout << "Total minutes (double): " << minAsDouble;

    return 0;
}
```

```
Enter the duration in seconds: 3670
Hours: 1
Minutes: 1
Seconds: 10
Total minutes (double): 61.1667
```

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Q3)

```cpp
#include <iostream>
#include <cmath>

int main(){
    float x = 2;
    float y = 2;
    float z = 3;


    x = x + (((-1 * y) + (4 * x)) / (2 * z)) - (((x * sqrt(y)) + 6) / 4);

    std::cout << x;
    return 0;
}
```

```
0.792893
```

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Q4)

```cpp
#include <iostream>

using namespace std;

int main(){
    int userNum1, userNum2;
    int difference, divisionResult, remainder;

    cout << "Enter the first number: ";
    cin >> userNum1;

    cout << "Enter the second number: ";
    cin >> userNum2;

    //Calculates if num1 can divide num2
    if (userNum2 != 0){
        if (userNum1/userNum2 != 0 && userNum1%userNum2 == 0) {
            cout << userNum1 << " is divisible by " << userNum2 << "." <<endl;
        }
        else {
            cout << userNum1 << " is not divisible by " << userNum2 << "." <<endl;
        }
    }
    else {
        cout << "Cannot divide by 0. " << userNum1 << " is not divisible by " << userNum2 << "." <<endl;
    }

    //Calculates if num2 can divide num1
    if(userNum1 != 0){
        if (userNum2/userNum1 != 0 && userNum2%userNum1 == 0) {
            cout << userNum2 << " is divisible by " << userNum1 << "." <<endl;
        }
        else {
            cout << userNum2 << " is not divisible by " << userNum1 << "." <<endl;
        }
    }
    else {
        cout << "Cannot divide by 0. " << userNum2 << " is not divisible by " << userNum1 << "." <<endl;
    }

    //Absolute difference
    difference = userNum1 - userNum2;

    if (difference < 0){
        difference *= -1;
    }

    cout << "The absolute difference between the two numbers is " << difference << endl;
```

```cpp

    //Divide num1 by num2 and show remainder
    if (userNum2 != 0){
        divisionResult = userNum1 / userNum2;
        remainder =  userNum1 % userNum2;

        cout << "The result of dividing " << userNum1 << " by " << userNum2 << " is " << divisionResult << " with a remainder of " << remainder <<endl;
    }
    else {
        cout << "Cannot divide by 0. " << userNum1 << " is not divisible by " << userNum2 << " and has no remainder." <<endl;
    }

    return 0;
}
```

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```
Enter the first number: 15
Enter the second number: 6
15 is not divisible by 6.
6 is not divisible by 15.
The absolute difference between the two numbers is 9
The result of dividing 15 by 6 is 2 with a remainder of 3
```

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Q5)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    double userIncome;
    double tax, finalIncome;

    cout << "Enter your annual salary: ";
    cin >> userIncome;

    // 10k
    if (userIncome < 10000){
        tax = 0;

        finalIncome = userIncome;
    }

    // 10 - 20k
    else if (userIncome >= 10000 && userIncome <= 20000){
        tax = (userIncome - 10000) * 0.1;
        tax -= 500;

        if (tax < 0){
            tax = 0;
        }

        finalIncome = userIncome - tax;
    }


    // 20 - 50k
    else if (userIncome > 20000 && userIncome <= 50000){
        tax = ((userIncome - 20000) * 0.2) + 1000;
        tax -= 1000;

        if (tax < 0){
            tax = 0;
        }

        finalIncome = userIncome - tax;
    }

```

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```
    // 50 - 100k
    else if (userIncome > 50000 && userIncome <= 100000){
        tax = ((userIncome - 50000) * 0.3) + 7000;
        tax -= 500;

        if (tax < 0){
            tax = 0;
        }

        finalIncome = userIncome - tax;
    }

    // 100k +
    else{
        tax = ((userIncome - 100000) * 0.4) + 22000;

        finalIncome = userIncome - tax;
    }

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    cout << "Salary: $" << userIncome << endl;
    cout << "Tax: $" << tax << endl;
    cout << "Salaty after tax: $" << finalIncome << endl;


    return 0;
}
```

```
Enter your annual salary: 10000
Salary: $10000.00
Tax: $0.00
Salaty after tax: $10000.00
```

```
Enter your annual salary: 25300
Salary: $25300.00
Tax: $1060.00
Salaty after tax: $24240.00
```

```
Enter your annual salary: 155000
Salary: $155000.00
Tax: $44000.00
Salaty after tax: $111000.00
```

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Q6)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    double weight, length, width, height;
    double dimensionalWeight, finalWeight, cost;

    cout << "Enter the package weight in kilograms: ";
    cin >> weight;

    cout << "Enter the package dimensions (length, width, height) in centimeters: ";
    cin >> length;
    cin >> width;
    cin >> height;

    dimensionalWeight = (length * width * height);
    dimensionalWeight /= 5000;

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    cout << "The dimensional weight is " << dimensionalWeight << " kg." << endl;

    if (dimensionalWeight > weight) {
        finalWeight = dimensionalWeight;
    }
    else {
        finalWeight = weight;
    }

    if (finalWeight <= 2){
        cost = 5;
    }
    else if (finalWeight > 2 && finalWeight <= 10) {
        cost = 10;
    }
    else if (finalWeight > 10 && finalWeight <= 20) {
        cost = 20;
    }
    else {
        cost = -1;
    }

    if (cost > 0){
        cout << "The shipping cost is $" << cost;
    }
```

```cpp
    else {
        cout << "The package is too heavy to be shipped.";
    }

    return 0;
}
```

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```
Enter the package weight in kilograms: 3
Enter the package dimensions (length, width, height) in centimeters: 50 40 30
The dimensional weight is 12.00 kg.
The shipping cost is $20.00
```

```
Enter the package weight in kilograms: 1.5
Enter the package dimensions (length, width, height) in centimeters: 20 10 5
The dimensional weight is 0.20 kg.
The shipping cost is $5.00
```

```
Enter the package weight in kilograms: 25
Enter the package dimensions (length, width, height) in centimeters: 60 50 40
The dimensional weight is 24.00 kg.
The package is too heavy to be shipped.
```

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Q7)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    int optionEntered;
    double userValue;
    double convertedValue;

    cout << "Enter the value you want to convert: ";
    cin >> userValue;

    cout << "Select the conversion:" << endl;
    cout << "1. Kilometers to Miles" << endl;
    cout << "2. Pounds to Kilograms" << endl;
    cout << "3. Celcius to Fahrenheit" << endl;
    cout << "4. Liters to Gallons" << endl;
    cout << "Enter your choice (1-4): ";
    cin >> optionEntered;

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    switch (optionEntered){
        case 1:
            convertedValue = userValue * 0.621371;
            cout << userValue << " kilometers " << "is equal to " << convertedValue << " miles.";
            break;
        case 2:
            convertedValue = userValue * 0.453592;
            cout << userValue << " pounds " << "is equal to " << convertedValue << " kilograms.";
            break;
        case 3:
            convertedValue = (userValue * 9 / 5) + 32;
            cout << userValue << " celcius " << "is equal to " << convertedValue << " fahrenheit.";
            break;
        case 4:
            convertedValue = userValue * 0.264172;
            cout << userValue << " liters " << "is equal to " << convertedValue << " gallons.";
            break;
        default:
            cout << "Invalid selection.";
            break;
    }

    return 0;
}
```

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```
Enter the value you want to convert: 100
Select the conversion:
1. Kilometers to Miles
2. Pounds to Kilograms
3. Celcius to Fahrenheit
4. Liters to Gallons
Enter your choice (1-4): 1
100.00 kilometers is equal to 62.14 miles.
```

```
Enter the value you want to convert: 50
Select the conversion:
1. Kilometers to Miles
2. Pounds to Kilograms
3. Celcius to Fahrenheit
4. Liters to Gallons
Enter your choice (1-4): 5
Invalid selection.
```

COEN 243 – Assignment 1

I/O stream and control statements (variables, operators, `if-else`, `switch-case`)

Deadline: Jan. 29th
Type: Individual Assignment
Weight: 4%

Submission instructions

- Create a `.cpp` file for each question.
- Create a single `.pdf` containing screenshots of **all programs** and their **results**.
- Put the PDF and all `.cpp` files in **one folder**.
- Name the folder: `lastname_studentid`.
- Compress the folder (ZIP or similar) and name the archive: `lastname_studentid`.
- Submit the archive on **Moodle**.
- Do **not** submit executable files.

Your Information

First Name:
Last Name:
Student ID:
Grade:

Expectations of originality:
Paste the signed form here or submit it as a separate file with this assignment.

Questions

1. **(10 marks)** Write a C++ program that uses **only** variables, `cin`, `cout`, and arithmetic operators (**no** `if`, `switch`, loops, or any branching).

   The program must:

   1) Ask the user to enter an **integer** number of items (`int`).
   2) Ask the user to enter the **price per item** (`double`).
   3) Ask the user to enter the **tax rate in percent** (`double`).
   4) Compute and print:
      - Subtotal = items × price
      - Tax amount = subtotal × (tax% / 100)
      - Total = subtotal + tax amount

The output should look like this:

```
*************************************************************************
Enter the number of items: 3
Enter the price per item: 20
Enter the tax rate (%): 15
Subtotal: $60
Tax: $9.00
Total: $69
*************************************************************************
```

2. **(10 marks)**Write a C++ program that uses **only** variables, `cin`, `cout`, and arithmetic operators (**no** `if`, `switch`, loops, or any branching).

   The program must:

   1) Ask the user to enter a duration in **seconds** (`int`).
   2) Convert it to **hours**, **minutes**, and **seconds** using integer division and modulo.
   3) Also compute the same duration in **minutes as a double** (e.g., 3670 seconds = 61.1667 minutes).

The output should look like this:

```
*************************************************************************
Enter the duration in seconds: 3670
Hours: 1
Minutes: 1
Seconds: 10
Total minutes (double): 61.1667
*************************************************************************
```

3. **(10 marks)** Write a C++ program to calculate the following formula:

$$x = x + \frac{-y + 4x}{2z} - \frac{x \cdot \sqrt{y} + 6}{4}$$

   using the initial values $x = 2$, $y = 2$, and $z = 3$.

   Note: Use `#include <cmath>` for `sqrt`. Make sure you use floating-point arithmetic.

4. **(15 marks)** Write a C++ program that asks the user to input two integers. The program will then:

   1) Print whether the first number is divisible by the second number or not.
   2) Print whether the second number is divisible by the first number or not.
   3) Compute and display the absolute difference between the two numbers.
   4) Compute and display the result of dividing the first number by the second number and the remainder.

Important: Your program must handle division by zero (if the user enters 0 as a divisor) by printing a clear message instead of performing the division/modulo.

The output should look like this:

```
****************************************
Enter the first number: 15
Enter the second number: 6
15 is not divisible by 6.
6 is not divisible by 15.
The absolute difference between the two numbers is 9.
The result of dividing 15 by 6 is 2 with a remainder of 3.
****************************************
```

5. **(35 marks)** You are tasked with building a **tax bracket calculator** that computes the tax based on the user's annual salary. The tax system is progressive, meaning different portions of the salary are taxed at different rates. Additionally, there are deductions that vary based on the salary range.

   Task: Write a C++ program that:

   (a) Asks the user to input their annual salary.

   (b) Applies the following tax brackets:

   - If the salary is **less than** $10,000, there is **no tax**.
   - If the salary is **between** $10,000 and $20,000 (inclusive), tax is **10% of the salary above $10,000** with a $500 deduction.
   - If the salary is **between** $20,001 and $50,000 (inclusive), tax is $1,000 for the first $20,000 plus **20% of the amount above $20,000**, with a $1,000 deduction.
   - If the salary is **between** $50,001 and $100,000 (inclusive), tax is $7,000 for the first $50,000 plus **30% of the amount above $50,000**, with a $500 deduction.
   - If the salary is **above** $100,000, tax is $22,000 for the first $100,000 plus **40% of the amount above $100,000**, with **no deduction**.

   (c) Ensure that the tax is **never negative** (e.g., when deductions exceed the calculated tax, tax is 0$).

   (d) Print the user's salary, the calculated tax, and the salary after tax.

The output should look like this:

```
*****************************************************************************

Example1:
Enter your annual salary: 10000
Salary: $10000.00
Tax: $0.00
Salary after tax: $10000.00

Example2:
Enter your annual salary: 25300
Salary: $25300.00
Tax: $1060.00
Salary after tax: $24240.00

Example3:
Enter your annual salary: 155000
Salary: $155000.00
Tax: $44000.00
Salary after tax: $111000.00

*****************************************************************************
```

6. **(20 marks)** Write a C++ program that calculates the shipping cost for a package based on its weight and dimensions (length, width, and height). The program will:

   1) Ask the user to input the package weight (in kilograms) and the package dimensions (length, width, and height in centimeters).
   2) Compute the **dimensional weight** as:

$$\text{dimensional weight} = \frac{\text{volume}}{5000} \quad \text{where volume} = \text{length} \times \text{width} \times \text{height}.$$

   3) Charge the user based on the **greater** of the actual weight and the dimensional weight:
      - If the greater weight is ≤ 2 kg, the cost is \$5.
      - If the greater weight is > 2 kg and ≤ 10 kg, the cost is \$10.
      - If the greater weight is > 10 kg and ≤ 20 kg, the cost is \$20.
      - If the greater weight is > 20 kg, the package is too heavy and cannot be shipped.

The output should look like this:

```
*************************************************************************
Enter the package weight in kilograms: 3
Enter the package dimensions (length, width, height) in centimeters: 50 40 30
The dimensional weight is 12.00 kg.
The shipping cost is $20.

Enter the package weight in kilograms: 1.5
Enter the package dimensions (length, width, height) in centimeters: 20 10 5
The dimensional weight is 0.20 kg.
The shipping cost is $5.

Enter the package weight in kilograms: 25
Enter the package dimensions (length, width, height) in centimeters: 60 50 40
The dimensional weight is 24.00 kg.
The package is too heavy to be shipped.
*************************************************************************
```

7. **(20 marks)** Write a C++ program that converts a given value from one unit of measurement to another. The program will prompt the user to enter a value and then select one of the following conversion options using a number:

   1) Convert from Kilometers to Miles (1 kilometer = 0.621371 miles)
   2) Convert from Pounds to Kilograms (1 pound = 0.453592 kilograms)
   3) Convert from Celsius to Fahrenheit (Fahrenheit = Celsius $\times 9/5 + 32$)
   4) Convert from Liters to Gallons (1 liter = 0.264172 gallons)

   Based on the user's selection, the program should perform the corresponding conversion and display the result. If the user enters an invalid option, the program should display an error message.

The output should look like this:

```
*************************************************************************
Enter the value you want to convert: 100
Select the conversion:
1. Kilometers to Miles
2. Pounds to Kilograms
3. Celsius to Fahrenheit
4. Liters to Gallons
Enter your choice (1-4): 1
100 kilometers is equal to 62.14 miles.
*************************************************************************
```

Invalid option example:

```
Enter the value you want to convert: 50
Select the conversion:
1. Kilometers to Miles
2. Pounds to Kilograms
3. Celsius to Fahrenheit
4. Liters to Gallons
Enter your choice (1-4): 5
Invalid selection.
```